

Problem Session 08: Lock Mechanisms

Peterson's Algorithm · LL/SC · MCS Lock

CPE 333 Operating Systems, 1/2026 — Group OS InW

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How We Evaluate

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Metrics from Lecture 8

Metric (Lecture 8)	Question	What we measure
Mutual exclusion	Is only one thread inside at a time?	Shared counter must equal total acquisitions: lost = total – counter = 0
Absence of deadlock	Does some waiting thread always get in?	Every run must finish when the time is up
Fairness	Does every thread get a fair chance?	Fewest ÷ most acquisitions per thread (1.0 = equal, 0 = starved)
Performance	How much time does the lock cost?	Acquisitions per second, with 1 to 8 threads

The Benchmark Loop

- All threads start together at a barrier
- Loop until main raises stop after 1 s
- counter is a plain variable: two threads inside together lose an update
- Each thread counts its own acquisitions (n)
- Same loop for every lock: only lock_acquire / lock_release change

```
static void *worker(void *arg)
{
    int id = (int)(long)arg;
    long n = 0;

    pthread_barrier_wait(&start);
    while (!stop) {
        lock_acquire(id);
        counter = counter + 1;           /* critical section */
        lock_release(id);
        n++;
    }
    acquired[id] = n;
    return NULL;
}
```

Reading One Run

```
--- tas threads=4 all-cpus run 1 ---  
$ bin/tas 4 1  
lock=tas threads=4 cpus=8  
  thread 0: 2246794 acquisitions  
  thread 1: 2607290 acquisitions  
  thread 2: 2315883 acquisitions  
  thread 3: 2362529 acquisitions  
total=9532496 counter=9532496 lost=0  
elapsed=1.000 s  throughput=9528568 acq/s  fairness(min/max)=0.862
```

- Mutual exclusion: total 9,532,496 = counter
9,532,496 → lost = 0
- Absence of deadlock: the run finished
- Fairness: $2,246,794 \div 2,607,290 = 0.862$
- Performance: 9.5 M acquisitions per second

Experiment Setup

Setting	Values
Locks	Peterson, LL/SC, MCS
References (Lecture 8)	TAS spin lock, pthread_mutex
Threads	1, 2, 4, 8 (Peterson: 1, 2)
CPUs	all 8, and one (taskset -c 0)
Runs	3 × 1 s per setting, median shown

- WSL2 Ubuntu, kernel 6.18.33.2-microsoft-standard-WSL2
- 8 CPUs
- gcc 15.2.0 -O2
- LL/SC on AArch64: qemu-aarch64 10.2.1
- One CPU shows what spinning costs when the lock holder cannot run

Peterson's Algorithm

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Peterson's Algorithm: Idea

- Pure software lock for two threads: only loads and stores
- $\text{flag}[i] = 1$ — “thread i wants to enter”
- $\text{turn} = \text{other}$ — “you go first”
- Wait while the other wants in and it is the other's turn
- The thread that wrote turn last waits

```
static volatile int flag[2];    /* flag[i] = 1: thread i wants in */
static volatile int turn;       /* whose turn it is to wait */

void lock_acquire(int self)
{
    int other = 1 - self;
    flag[self] = 1;             /* I want to enter ... */
    turn = other;               /* ... but you go first */
    FENCE();                   /* stores done before the loads */
    while (flag[other] == 1 && turn == other)
        ;                      /* spin: you want in, your turn */
}

void lock_release(int self)
{
    FENCE();
    flag[self] = 0;
}
```

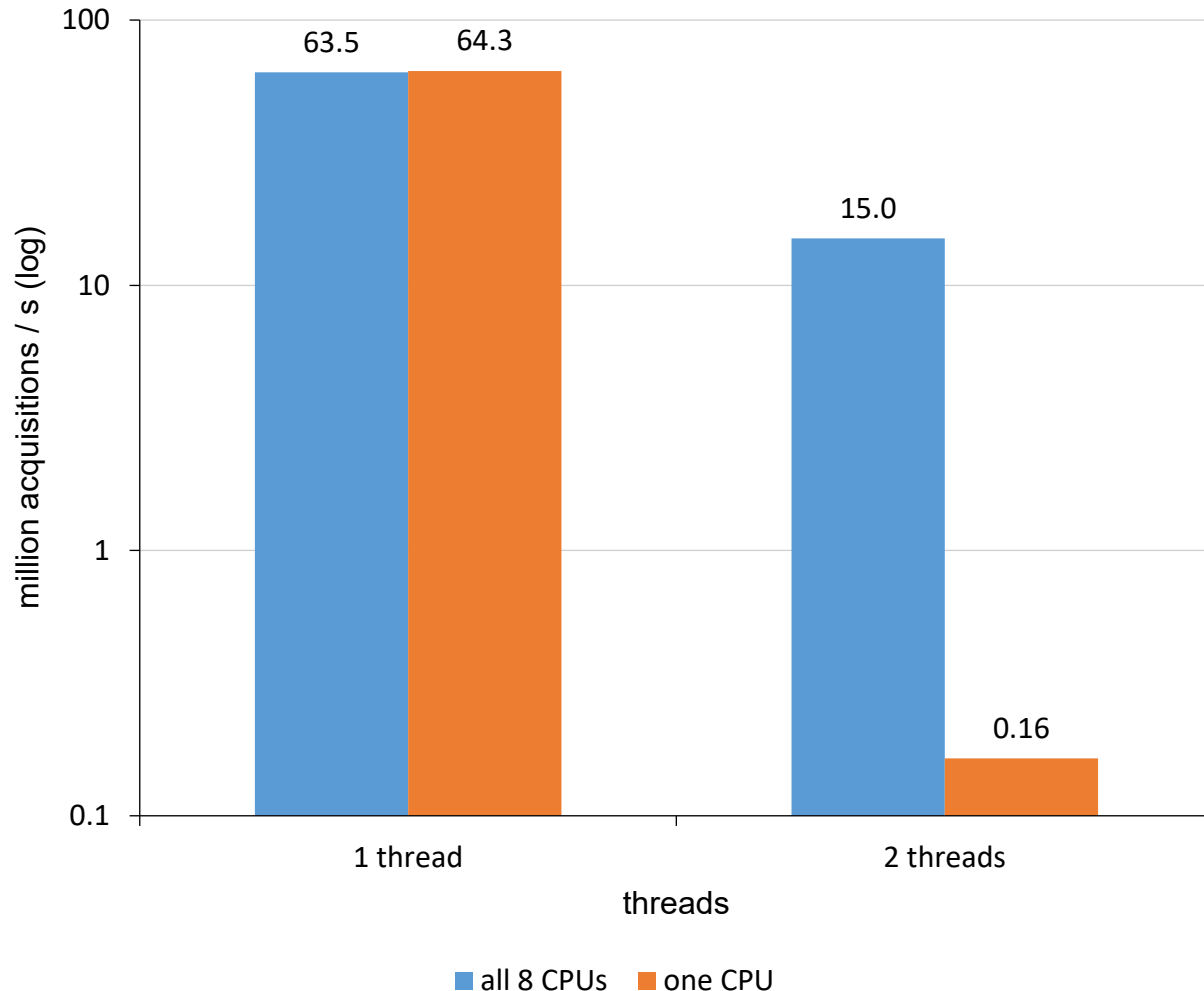
Peterson: Mutual Exclusion and Deadlock

- Mutual exclusion: if both were inside, each saw `turn == itself` — impossible, `turn` holds one value
- No deadlock: `turn` is 0 or 1, so one waiting thread always passes
- But x86 lets a load pass an earlier store (store buffer): both read `flag[other] == 0` and both enter

Build (2 threads)	CPUs	Runs with lost updates	Lost updates per run
no fence	all 8	5 / 5	233,108 – 813,401
no fence	one	0 / 3	0
with FENCE()	all 8	0 / 5	0

- `FENCE()` compiles to `lock orq $0x0,(%rsp)` — a full barrier: the stores drain before the loads

Peterson: Performance



- 1 thread: 63 M acq/s — no contention
- 2 threads, all CPUs: 15 M acq/s — flag and turn move between cores
- 2 threads, one CPU: 165 k acq/s — the waiter spins until the scheduler switches
- Busy-waiting costs a whole time slice when the other thread is not running

Peterson: Fairness and Limitations

- All CPUs, 2 threads: fairness 0.99 — turn gives way to the other thread, so a waiter enters after at most one entry by the other (bounded waiting)
- One CPU, 2 threads: fairness 0.001 (run 1: 126 vs 93,264) — with spinning, the scheduler decides
- Limitations:
 - Two threads only (N threads need a tournament of Peterson locks or the filter lock)
 - Needs a memory fence on modern CPUs — the textbook version is broken on x86
 - Spins instead of sleeping

Load-Linked / Store-Conditional

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LL/SC: Idea

- LoadLinked(p): load *p and watch the address
- StoreConditional(p, v): store only if nobody stored to p since the LoadLinked
- Lock: spin on LL until 0, then SC 1; if SC fails, start over
- MIPS, Alpha, PowerPC, ARM (AArch64: LDXR/STXR)
- x86 has no LL/SC — it offers CAS (lock cmpxchg)

```
static inline int LoadLinked(volatile int *p)
{
    int v;
    __asm__ volatile("ldaxr %w0, [%1]"
                     : "=r"(v) : "r"(p) : "memory");
    return v;
}

static inline int StoreConditional(volatile int *p, int v)
{
    int fail;
    __asm__ volatile("stxr %w0, %w2, [%1]"
                     : "=r"(fail) : "r"(p), "r"(v) : "memory");
    return !fail;
}

void lock_acquire(int id)
{
    (void)id;
    while (1) {
        while (LoadLinked(&flag))
            ; /* spin until it is zero */
        if (StoreConditional(&flag, 1))
            return; /* no store in between: ours */
        /* else another thread won: try again */
    }
}
```

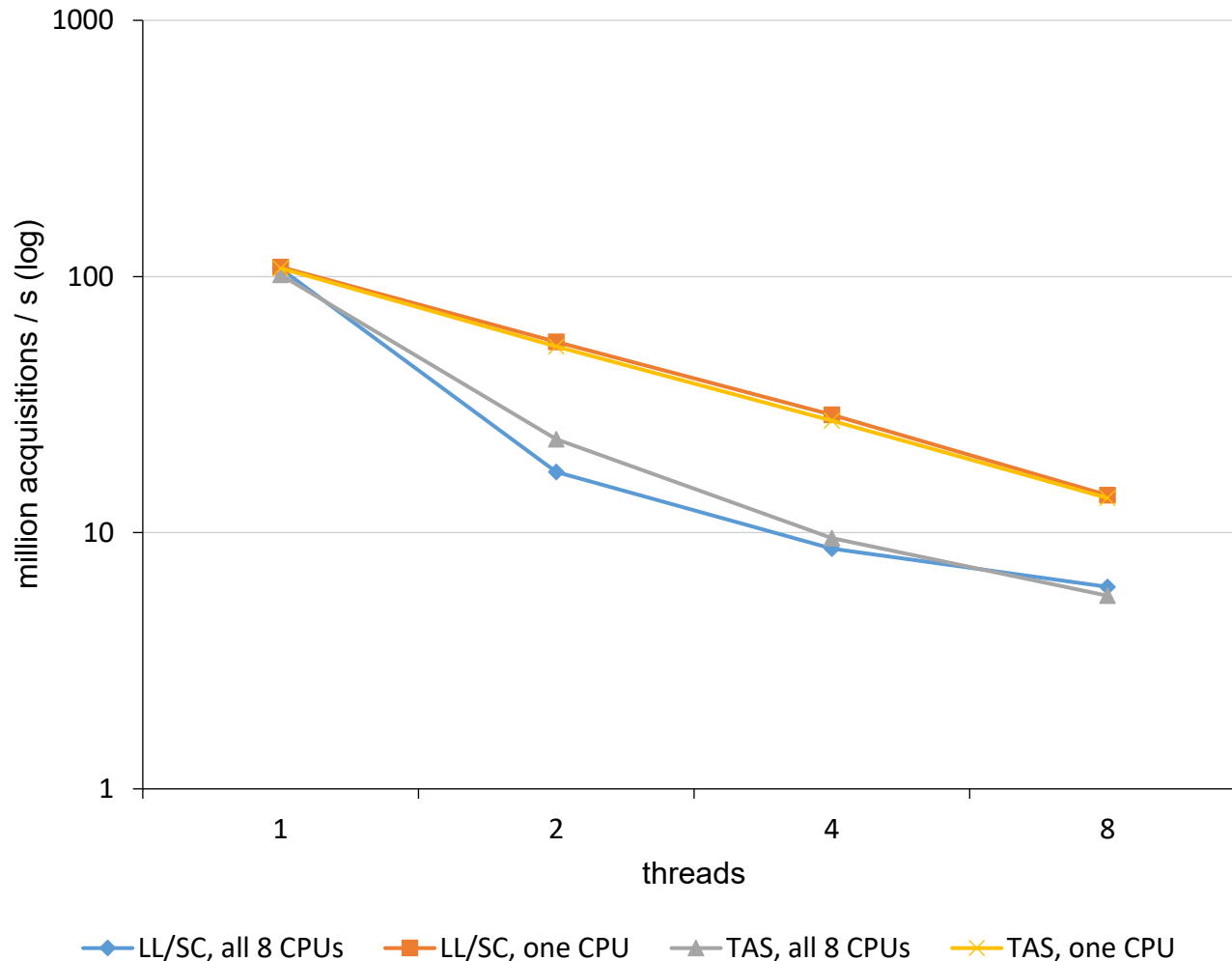
LL/SC: Mutual Exclusion and Deadlock

- Mutual exclusion: two threads may both LL a 0, but the first SC writes the flag, so the second SC fails
- No deadlock: an SC fails only if another store got in (that thread now holds the lock) or spuriously (e.g. an interrupt) — the loop retries
- AArch64 build under qemu-aarch64 10.2.1, 4 threads: lost = 0 in 3 / 3 runs (5,252,862+ acquisitions each)

```
0000000000400b8c <lock_acquire>:  
adrp x1, 4b1000  
add x1, x1, #0xb40  
mov w2, #0x1  
nop  
nop  
ldaxr w0, [x1]  
cbnz w0, 400ba0 <lock_acquire+0x14>  
stxr w0, w2, [x1]  
cbnz w0, 400ba0 <lock_acquire+0x14>  
ret
```

ldaxr = Load-Linked, stxr = Store-Conditional (writes 0 to w0 on success)

LL/SC: Performance



- Timed on x86 with CAS standing in for LL/SC (qemu speed is meaningless)
- All CPUs: 107 M acq/s with 1 thread → 6.1 M with 8 — every try pulls the flag's cache line
- One CPU, 8 threads: 14 M acq/s
- Waiters only read (LL) until the flag looks free, then write once (SC)

LL/SC: Fairness and Limitations

- No queue: the first SC to land wins — fairness at 8 threads 0.75 (all CPUs), 0.00 (one CPU; run 1: the unluckiest thread got 1 of 14,638,997)
- A thread preempted while holding the lock makes every other thread spin out its time slice
- Limitations:
 - Starvation is possible; spins instead of sleeping
 - Hardware-specific: not on x86; SC may fail spuriously
 - Little may run between LL and SC, or the reservation is lost
- Advantage over CAS: SC fails on any store, even one that writes back the same value (no ABA problem)

MCS Lock

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MCS Lock: Idea

- Mellor-Crummey and Scott, 1991; Dijkstra Prize 2006
- Waiters form a queue: a linked list of nodes, tail points to the last
- Join: one atomic exchange on tail, then link in behind the previous node
- Each thread spins on its OWN node; the one ahead clears it on release
- Linux's spinlock (qspinlock, since v4.2) is built on MCS

```
struct node {
    struct node *next;          /* the thread queued behind me */
    int locked;                 /* 1 = keep waiting */
} __attribute__((aligned(64))); /* one cache line per node */

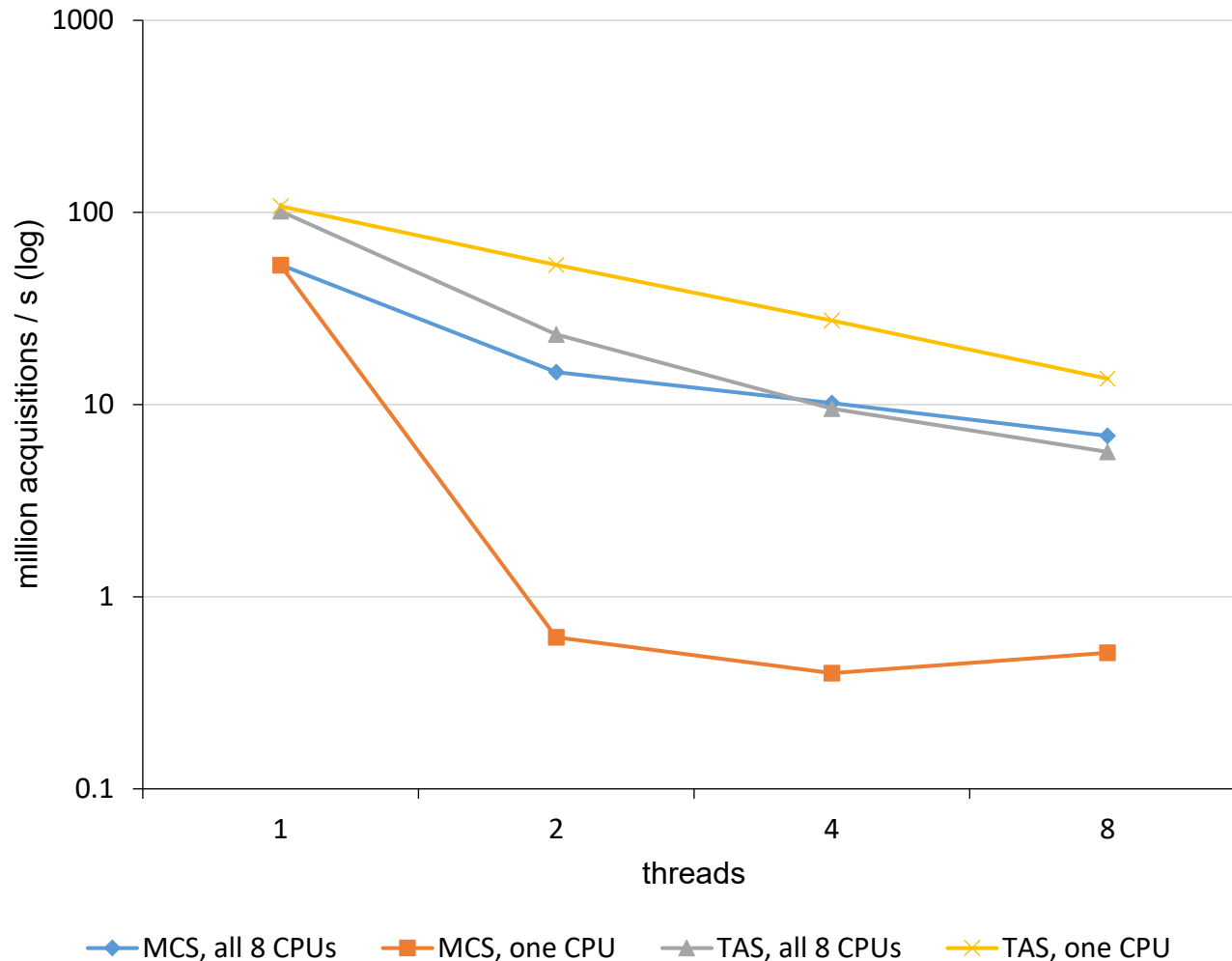
void lock_acquire(int id)
{
    struct node *me = &qnode[id];
    me->next = NULL;
    me->locked = 1;
    struct node *prev = __atomic_exchange_n(&tail, me, __ATOMIC_ACQ_REL);
    if (prev != NULL) {         /* queue not empty: */
        STORE(&prev->next, me); /* link in behind prev */
        while (LOAD(&me->locked))
            ;                    /* spin on MY node only */
    }
}
```

MCS: Mutual Exclusion and Deadlock

- Mutual exclusion: the exchange gives every thread one place in one queue; only the head is inside, and it wakes exactly one node
- No deadlock: the holder always hands over — to the next node, or resets tail when nobody waits
- release() waits for a waiter that has swapped tail but not linked in yet
- Measured: all 24 runs (1–8 threads, all CPUs and one CPU) finished with lost = 0

```
void lock_release(int id)
{
    struct node *me = &qnode[id];
    struct node *next = LOAD(&me->next);
    if (next == NULL) {
        struct node *expect = me;
        if (__atomic_compare_exchange_n(&tail, &expect, NULL, 0,
                                         __ATOMIC_RELEASE, __ATOMIC_RELAXED))
            return;                /* nobody waiting: lock is free */
        while ((next = LOAD(&me->next)) == NULL)
            ;                      /* a waiter is still linking in */
    }
    STORE(&next->locked, 0);      /* hand the lock to the next thread */
}
```

MCS: Performance



- Each waiter spins on its own cache line; a handover touches one line
- All CPUs, 8 threads: 6.9 M acq/s (TAS: 5.7 M)
- One CPU, 8 threads: 511 k acq/s — the next in the queue is not running
- In the kernel, spinlocks run with preemption disabled, so a queued waiter is never descheduled

MCS: Fairness and Limitations

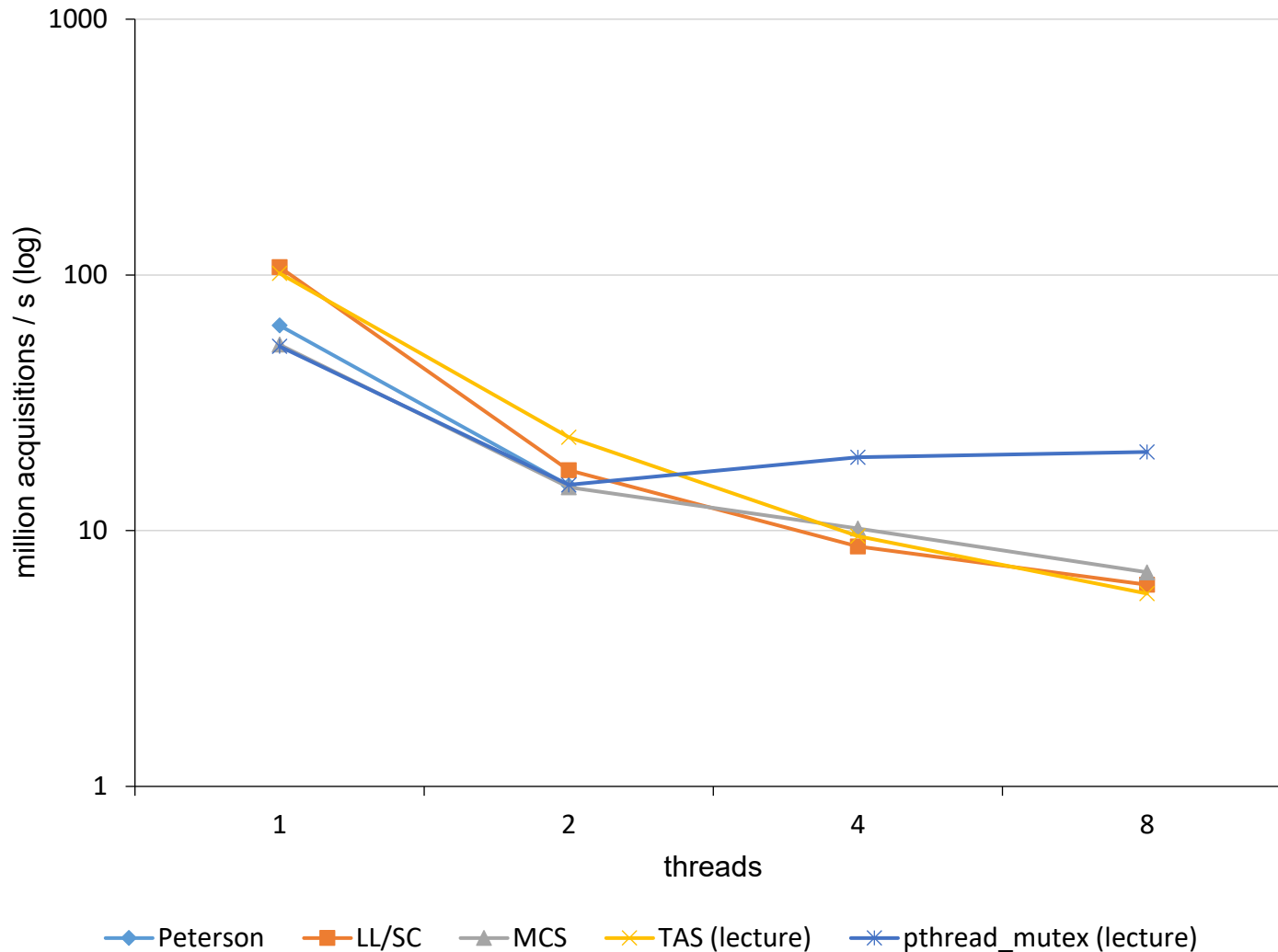
- First-come, first-served through the queue: fairness 1.00 / 1.00 / 1.00 at 2/4/8 threads (all CPUs) — the fairest lock at 8 threads
- Limitations:
 - Needs an atomic exchange (and CAS for release) from the hardware
 - One queue node per waiting thread, passed in or kept per thread
 - A preempted waiter stalls everyone behind it — the kernel disables preemption
 - On one CPU here: 511 k acq/s at 8 threads, fairness 0.00

Comparison: Throughput

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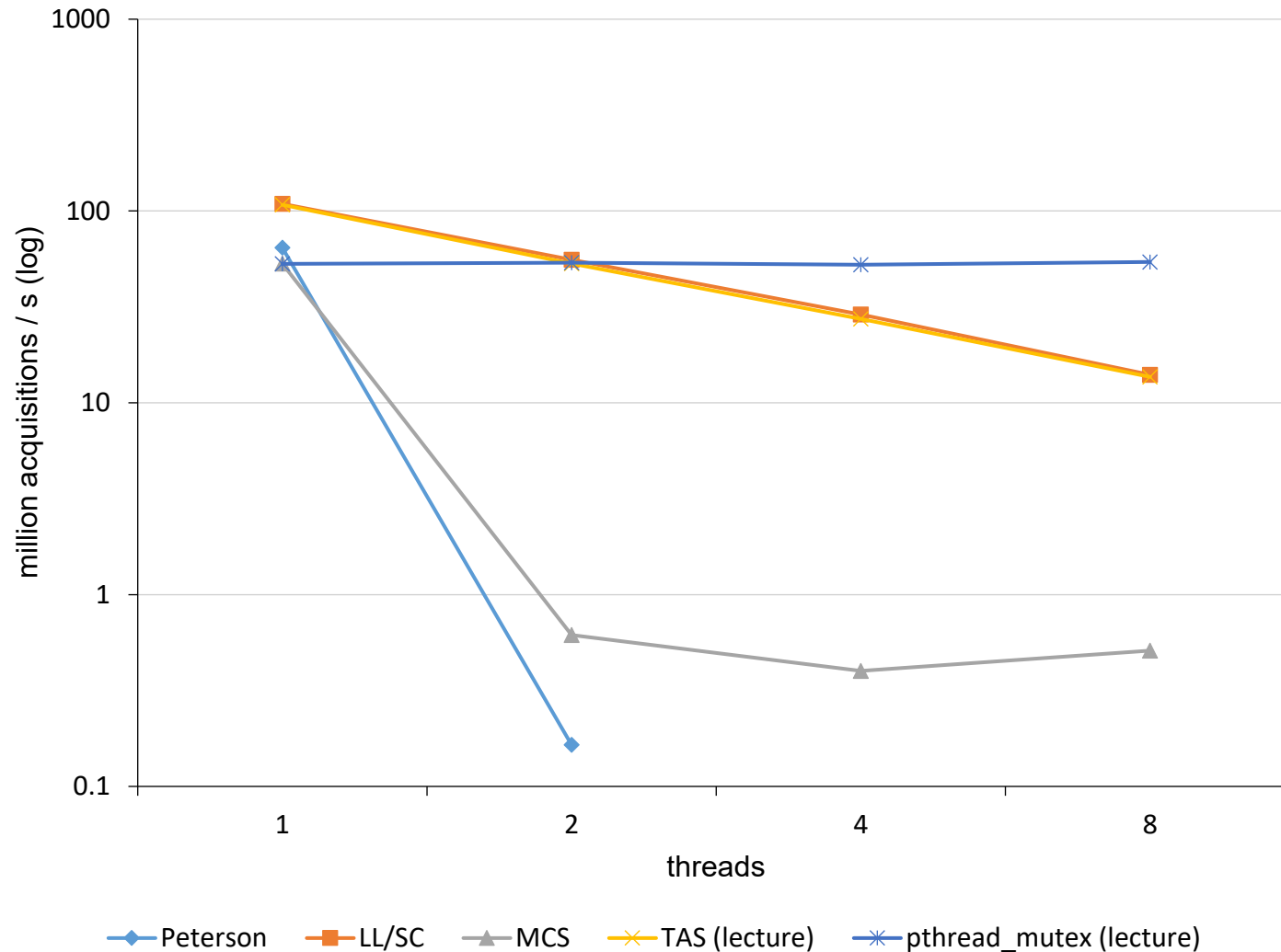
Worawut Sereethai

Throughput on All 8 CPUs



- 1 thread, no contention: LL/SC 107 M and TAS 101 M acq/s are fastest
- More threads: every spin lock slows down as threads fight over cache lines
- 8 threads: MCS 6.9 M, LL/SC 6.1 M, TAS 5.7 M acq/s
- Fastest at 8 threads: pthread_mutex, 20 M acq/s

Throughput on One CPU



- TAS / LL/SC stay fast (14 M / 14 M at 8 threads), but one thread holds on while the others starve
- Peterson and MCS hand the lock to a thread that is not running: 165 k (Peterson, 2 thr), 511 k (MCS, 8 thr)
- **pthread_mutex sleeps instead of spinning: 53 M → 54 M acq/s from 1 to 8 threads**

Comparison: Summary

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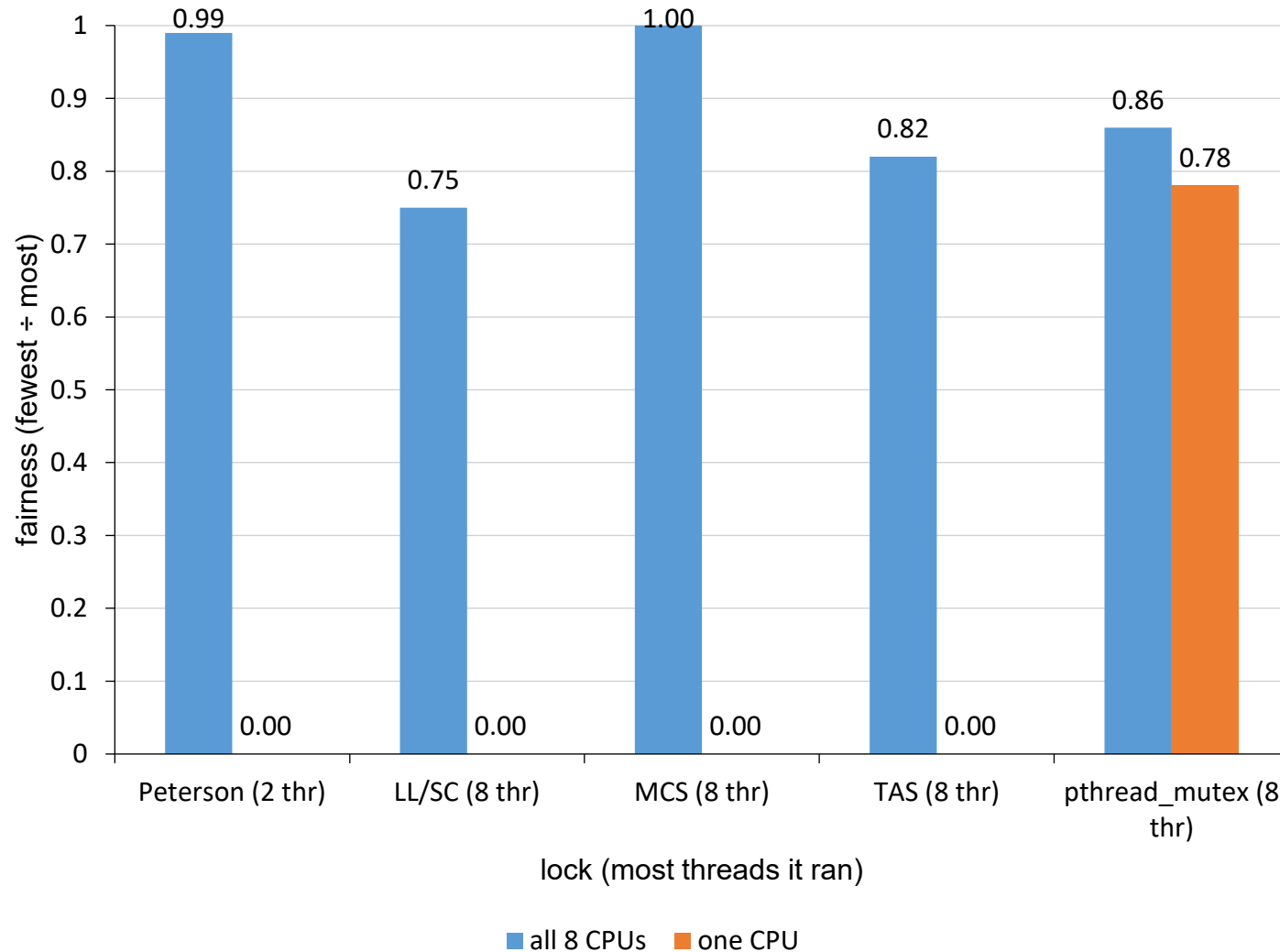
Chanon Lhumsa-ard

Correctness: Mutual Exclusion and Deadlock

Lock	Runs	Lost updates	Runs finished	Also tested
Peterson	17	0	17 / 17	no fence: lost updates in 5 / 5 runs
LL/SC	24	0	24 / 24	real ldaxr/stxr under qemu: lost = 0 in 3 / 3 runs
MCS	24	0	24 / 24	
TAS	24	0	24 / 24	
pthread_mutex	24	0	24 / 24	

- Mutual exclusion: lost = total acquisitions – counter, summed over all runs
- Peterson is correct on x86 only with the fence

Fairness



- All CPUs: MCS 1.00 is the fairest at 8 threads — its queue is first-come, first-served
- TAS 0.82, LL/SC 0.75: whoever wins the race gets in
- One CPU: every spin lock lets a thread starve (the bars near 0) — the scheduler decides who gets in
- Only pthread_mutex stays fair on one CPU: 0.78

Summary

Lock	Needs	Mutual excl.	Deadlock-free	Fairness	acq/s, all 8 CPUs	acq/s, one CPU
Peterson (2 thr)	loads/stores + fence	yes	yes	0.99	15 M	165 k
LL/SC (8 thr)	LL/SC instructions	yes	yes	0.75	6.1 M	14 M
MCS (8 thr)	atomic exchange + CAS	yes	yes	1.00	6.9 M	511 k
TAS (8 thr)	test-and-set	yes	yes	0.82	5.7 M	14 M
pthread_mutex (8 thr)	atomics + futex	yes	yes	0.86	20 M	54 M

- Fairness = $\text{fewest} \div \text{most acquisitions per thread}$; medians of 3 runs of 1 s
- Peterson gives mutual exclusion on x86 only with the fence

Conclusion

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Conclusion

- All three give mutual exclusion and no deadlock — Peterson only with a fence on modern CPUs
- Peterson: software only and fair (bounded waiting), but two threads only
- Hardware LL/SC: one short loop, fast, but no fairness and not on x86
- MCS: a queue gives first-come, first-served for any number of threads; Linux builds on it
- Spinning on one CPU wastes time slices; sleeping locks (futex, pthread_mutex) avoid it

References

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- CPE 333 Lecture 8: Locks Mechanism
- G. L. Peterson, “Myths About the Mutual Exclusion Problem”, Information Processing Letters 12(3), 1981
- J. M. Mellor-Crummey, M. L. Scott, “Algorithms for Scalable Synchronization on Shared-Memory Multiprocessors”, ACM TOCS 9(1), 1991
- Linux kernel source: kernel/locking/qspinlock.c
- Arm Architecture Reference Manual for A-profile architecture: LDAXR, STXR
- Code and raw results: PS08/src, PS08/results